



KENYATTA UNIVERSITY
UNIVERSITY EXAMINATIONS 2011/2012

INSTITUTION BASED PROGRAMME

DECEMBER SESSION EXAMINATION FOR THE DEGREE OF MASTER OF EDUCATION

ECT 534: APPLICATION OF MATHEMATICS IN DAILY LIFE

DATE: Friday 4th May, 2012

TIME: 9.00 a.m. – 12.00 p.m.

INSTRUCTIONS: Answer *THREE* questions including Question 1.

1.
 - (a) State an objective of secondary school mathematics and relate it to how it influences daily life at individual, school and society level. [6 marks]
 - (b) Besides being an essential element of communication, mathematics is known for its utility. Discuss this value of mathematics in our daily lives. [8 marks]
 - (c) Discuss how research in mathematics impacts on daily life. [8 marks]
 - (d) Describe the link between secondary school mathematics and physical education. [8 marks]
2.
 - (a) Describe how the introduction of calculators can enhance or limit student engagement in mathematics. [5 marks]
 - (b) What are some of the potential roles of a computer in the classroom? For each role, identify value and limitation. [10 marks]
3.
 - (a) Is mathematics dead or alive? Justify your response by use of examples.
 - (b) What is the role of mathematics in National development? Consider at least four different sectors of the national economy. [15 marks]
4. When students are asked about their views about mathematics, they state that “it is difficult” further probing always results in identification of *inappropriate learning environment*. Discuss this view. How would relate teachers knowledge and attitude towards mathematics to student learning. [15 marks]

5. (a) In your opinion, why people make mistakes in mathematics. How can teachers use non examples and student's errors to support learning of mathematics. [7 marks]
- (b) What does it mean to understand conceptually and how can we help pupils modify their understanding. Give examples from Kenyan school mathematics curriculum. [8 marks]

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